1 A student is investigating how the extension e of an elastic cord depends on the diameter d of the cord when a force is applied.

The student has a number of elastic cords of the same material with different diameters. The elastic cords have circular cross-sections. Each cord has an unstretched length of approximately 50 cm.

It is suggested that the extension e and the cross-sectional area A of the cord are related by the expression

$$E = \frac{FL}{Ae}$$

where E is the Young modulus of the material of the cord, F is the force applied and L is the unstretched length of the cord.

Design a laboratory experiment to test the relationship between e and d.
Explain how your results could be used to determine a value for E.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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